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Camera-Only, Continuous Real-Time Marathi Sign Language Recognition: A Systematic Literature Review of Hybrid 1D-CNN–BiLSTM–Temporal Attention Architectures

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Abstract

Background: Sign language is the primary mode of communication for a large part of the deaf and hard-of-hearing community in Maharashtra, India, yet there is very little research on automatic recognition of Marathi Sign Language (MSL). Most existing systems classify isolated signs from static images under controlled conditions. Such systems do not work well in real settings such as hospitals, classrooms and government offices, where signing is continuous. **Methods:** This paper presents a systematic literature review (SLR) of sign language recognition systems with respect to architecture, feature modality, datasets and real-time deployment. The review follows the PRISMA 2020 guideline. From 140 records retrieved between 28 July and 5 August 2026 from 8 digital libraries, 15 studies (mean quality score 9.0/10) were included after applying the inclusion and exclusion criteria and quality appraisal. **Findings:** Most of the work uses appearance-based convolutional neural network (CNN) pipelines. Hybrid CNN–LSTM and Transformer sequence models are increasingly used for sequence-level recognition, but they are computationally costly. None of the reviewed studies addresses continuous signing with idle states or ambient motion. Almost all studies use privately collected datasets, and no public benchmark exists for continuous MSL recognition. The review identifies four research gaps: computational cost, spatial bias and domain shift, temporal hallucination during idle periods, and scarcity of regional data. **Conclusion:** We propose a zero-pixel MediaPipe framework with wrist-anchored normalisation, a hybrid 1D-CNN + BiLSTM + multi-head temporal-attention classifier, a 30-frame rolling buffer with an explicit Idle class, and an 85% confidence debounce filter, together with a validation plan to test the design.

Keywords: Marathi Sign Language; Continuous Sign Language Recognition; MediaPipe Skeletal Tracking; BiLSTM with Temporal Attention; Real-Time Inference; Assistive Technology

1. Introduction

1.1 Background

Sign languages are complete natural languages with their own phonology, morphology and grammar [2]. For most deaf and hard-of-hearing people, a sign language is the first language they learn. The Rights of Persons with Disabilities Act, 2016 recognises sign language as a mode of communication in India [1], but interpreters are few. As a result, many deaf people in Maharashtra depend on relatives or on written notes when they visit hospitals, schools, banks and other public places. Marathi Sign Language (MSL) has a Devanagari-based manual alphabet and a regional lexicon used by Marathi-speaking deaf communities, yet it has received very little attention from the automatic recognition community.

Research on automatic sign language recognition has changed direction over time. Early vision-based systems treated recognition as the classification of one frame, or a short sequence of frames, into one of a few labels from a fingerspelled alphabet or a small set of isolated words. This isolated, appearance-based formulation is described in the survey literature [2] and is regarded as the historical baseline. Real signing is not isolated. Signs co-articulate with no clear word boundaries, and the hands return to a resting position between utterances. Continuous recognition therefore has to model variable-length sequences instead of classifying frames, and it has to decide where each sign starts and ends [3], [4]. Alignment-free training methods such as Connectionist Temporal Classification (CTC) were introduced to solve the boundary problem when frame-level labels are not available [5].

1.2 Motivation

This review is motivated by three observations. First, a recognition engine meant for real users must run on standard, inexpensive equipment such as a laptop or phone camera, without gloves or depth sensors. Pixel-based models need a lot of computing power and are sensitive to lighting, background, clothing and camera framing. Second, on-device landmark trackers have become much more capable. The MediaPipe framework and its hand and body-pose models give a compact set of 3D landmarks in real time on commodity hardware [6]–[8]. A zero-pixel approach discards appearance information completely, gives a smaller input and lower compute, and has an inherent privacy benefit because no face or background pixels are stored or sent. Third, regional languages have far fewer resources than widely used sign languages. Large public databases exist for American Sign Language (ASL) isolated words, such as WLASL [10], and for Indian Sign Language (ISL) isolated words, such as INCLUDE [9], but no comparable data covers MSL sentences.

This paper is part of the final-year project BhashaBridge at PCCoE, which aims to build an economical, camera-based MSL recognition engine. The review is designed to identify what the literature already answers, what it leaves open, and which design decisions the evidence supports.

1.3 Problem Statement

Prior sign language recognition studies concentrate on a small number of high-resource sign languages and usually define recognition as a static or isolated sign classification task in a laboratory setting. For regional languages such as MSL, there is no shared understanding of suitable feature representations, temporal architectures and inference strategies for continuous, camera-only recognition on edge hardware, nor of the failure modes that must be engineered against, namely domain shift and spurious outputs during idle periods.

1.4 Objectives

The review has four objectives, each aligned with a research question in Section 2:

1. To trace the evolution of architectures in sign language recognition, from CNNs to recurrent and Transformer-based models.
2. To compare RGB (appearance-based) and skeletal (landmark-based) feature extraction.
3. To characterise the limitations of datasets for regional sign languages, with a focus on MSL.
4. To study the constraints of real-time deployment and the hallucination problem in continuous inference.

1.5 Contributions of the SLR

This review provides (i) a synthesis of 15 recent studies, organised by modality, architecture and deployment scenario rather than as a study-by-study overview; (ii) a comparison of RGB and skeletal pipelines with a clear statement of the trade-off of each; (iii) a dataset analysis that separates static

alphabet resources from continuous, sentence-level corpora; (iv) four evidence-based research gaps; and (v) a reference design for a zero-pixel continuous MSL engine, with a validation agenda.

2. Research Questions and Objectives

Four research questions (RQs) were defined before screening. Each maps to one objective in Section 1.4.

RQ1 (Objective 1, Architectures). How do recognition architectures differ from CNNs to LSTM-based and Transformer-based models, and which are appropriate for a continuous sign sequence?

RQ2 (Objective 2, Feature extraction). What are the advantages and disadvantages of RGB-based and skeleton-based feature extraction for sign recognition?

RQ3 (Objective 3, Datasets). Which datasets and evaluation methods are available, and what are their weaknesses for regional languages, especially MSL?

RQ4 (Objective 4, Deployment). Which real-time limitations and failure modes, such as false predictions (hallucinations), are reported, and how are they addressed?

3. SLR Methodology

3.1 Review Protocol

The review follows the PRISMA 2020 reporting guideline [11] and the planning, conducting and reporting stages of Kitchenham and Charters [12]. The protocol was fixed before screening and covered the keywords, databases, eligibility criteria, selection stages, quality-assessment questions and data-extraction fields. The database searches were run between 28 July and 5 August 2026 (Appendix A).

3.2 Information Sources

Eight online sources were used: IEEE Xplore, Google Scholar (as a cross-check), SpringerLink, ScienceDirect, PubMed/PMC, Wiley Online Library, Nature (Scientific Reports) and Frontiers. The per-database queries, filters and record counts are given in Appendix A.

3.3 Search Strategy

Search terms were grouped into concept blocks: sign language and gesture recognition; bidirectional or continuous translation; and deep-learning methods. The general Boolean string used in IEEE Xplore, and then translated into the syntax of the other sources, was:

> ("sign language recognition" OR "gesture recognition") AND ("bidirectional translation" OR "sign-to-speech" OR "speech-to-sign") AND ("deep learning" OR "YOLO" OR "large language model")

Because this review focuses on skeleton-based continuous recognition, a second, focused string was also defined for Scopus and the ACM Digital Library for replication. It was not one of the searches executed and reported in Fig. 1 and Appendix A.

> TITLE-ABS-KEY(("sign language" OR "gesture recognition") AND ("continuous" OR "sentence-level" OR "real-time" OR "dynamic") AND ("skeleton" OR "landmark" OR "pose estimation" OR "MediaPipe" OR "keypoint") AND ("LSTM" OR "BiLSTM" OR "CNN" OR "transformer" OR "attention")) AND PUBYEAR > 2021 AND LANGUAGE(English)

The main search window was 2022–2026. Works outside this window that are foundational or that introduced benchmark datasets or tools are cited as supplementary literature for context and are not part of the PRISMA flow.

3.4 Inclusion Criteria

- **IC1:** Peer-reviewed journal or conference papers published in 2022–2026.
- **IC2:** Sign language recognition or translation using deep-learning or AI techniques.
- **IC3:** Camera-based input, or a camera-based alternative to specialised hardware.
- **IC4:** Low-resource languages (NLP) or system architectures for assistive communication, where applicable to the research questions.
- **IC5:** Full text available in English.

3.5 Exclusion Criteria

- **EC1:** Publications before 2022.
- **EC2:** Manual or statistical (non-AI) methods.
- **EC3:** Systems that need special external hardware, such as flex-sensor gloves, with no camera-based alternative.
- **EC4:** Non-peer-reviewed material or copies of earlier work that is not original.
- **EC5:** Lack of detail or scope in the methodology, or work beyond the four research questions.

3.6 Quality Assessment

Each included study was scored on five questions: QA1 (Is the objective clearly stated?), QA2 (Is the method described adequately?), QA3 (Is the data described adequately?), QA4 (Is the evaluation reported clearly and quantitatively?) and QA5 (Are the limitations discussed?). Each was marked Yes = 2, Partially = 1, No = 0, for a maximum of 10. Studies were rated High (8–10), Moderate (6–7) or Low (below 6); studies scoring below 6 were not to be included. The results are in Section 5.7.

4. Literature Search and Selection

The searches returned 140 records. After 18 duplicates were removed, 122 records were screened by title and abstract and 84 were excluded, mainly because they were not AI methods, relied on specialised hardware, or were published before 2022. Thirty-eight full texts were assessed against the eligibility criteria. Twenty-three were dropped because of insufficient methodological detail, purely statistical methods, or a scope beyond the four research questions. Fifteen studies (S1–S15) met all the inclusion criteria and form the final synthesis. The PRISMA flow is shown in Fig. 1.

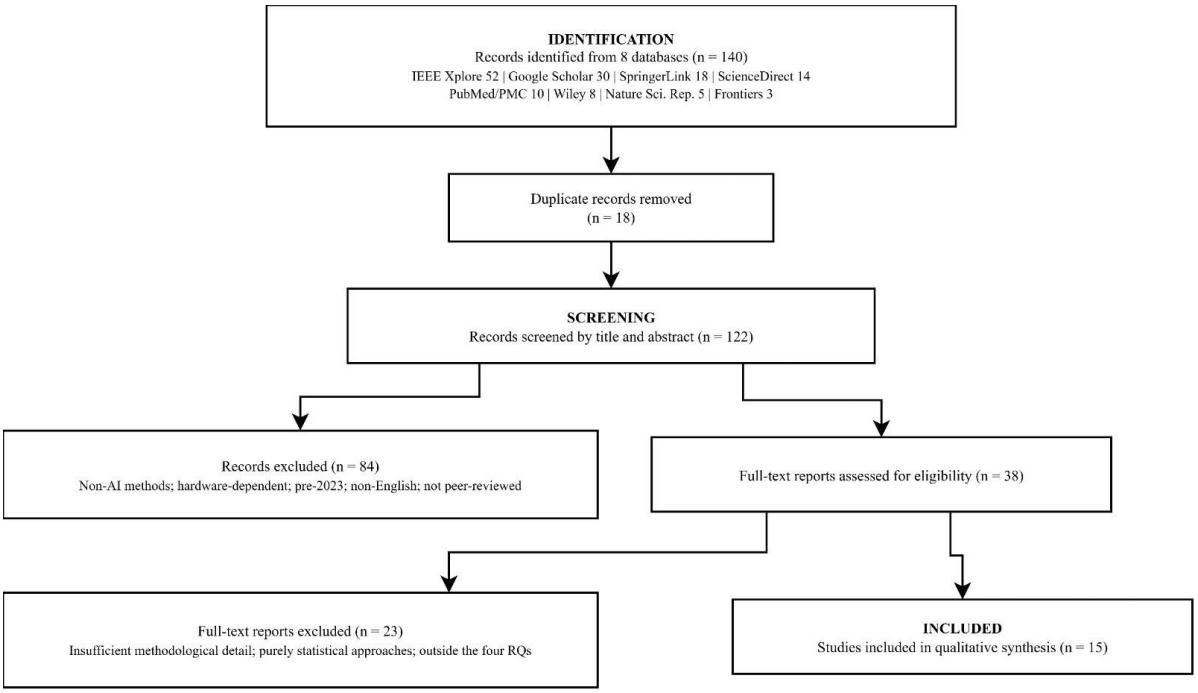


Fig. 1. PRISMA 2020 flow diagram of study identification, screening and inclusion.

5. Analysis and Synthesis of Selected Studies

5.1 Corpus Characteristics and Publication Trends

The screening produced three groups of primary studies (S1–S15), supported by wider domain literature ([3]–[10], [28]–[34], [46]–[60]), organised by dataset and input modality. The first group uses static image recognition to classify isolated alphabets, digits or hand postures from a single RGB frame. Tipare et al. [13] used a static 2D-CNN gesture classifier, Mahendran and Mahendran [14] a deep CNN for ISL alphanumerics, Patil et al. [15] a prototype ISL-to-speech deep neural network, and Dutta et al. [16] a Convolutional Vision Transformer (CvT). Sosa-Jiménez et al. [17] built a prototype for Mexican Sign Language for primary-care consultation. In the wider literature, Deshmukh et al. [46] applied a VGGNet transfer-learning model to static MSL characters, Wadhawan and Kumar [47] a deep CNN for static ISL signs, Katoch et al. [48] a SURF and bag-of-visual-words pipeline with a CNN, and Sharma et al. [49] a transfer-learning benchmark. Giri et al. [50] then proposed a static MSL classification system using MediaPipe keypoints.

The second group models multi-frame hand trajectories or landmark sequences: the DeepSign LSTM–GRU framework [51], a real-time dynamic ISL pipeline [52], a MediaPipe–LSTM network [53], a Transformer over MediaPipe landmarks [54], and a custom attention-enhanced LSTM for dynamic ISL [55], together with multimodal and bidirectional translation architectures [21]–[27], [31], [33]. The third group consists of continuous video recognition systems: Abeje et al. [18] (hybrid CNN–LSTM for Ethiopian Sign Language video), a video-based statistical-ML approach [19], a hybrid deep-learning and fuzzy-matching approach for Arabic Sign Language video [20], earlier statistical and deep approaches [3], [4], [28]–[30], and modern video architectures such as C2SLR [56], the Self-Emphasizing Network (SEN) [57], CorrNet [58], NLA-SLR [59] and SLRFormer [60].

Publication activity is concentrated in recent years: 2 of the 15 primary studies were published in 2022, 2 in 2023, 3 in 2024 and 8 in 2025 (53%). Eleven of the 15 appeared in IEEE venues, 2 in Springer venues and 2 in other journals. The increase in 2025 coincides with growing interest in combining recognition with language models and speech components.

5.2 Summary of Key Literature

The reviewed literature is summarised in three tables. Table I covers studies on static image datasets. Table II covers studies on dynamic image sequences and skeletal gesture trajectories. Table III covers studies on word-level and continuous video streams.

TABLE I. SUMMARY OF LITERATURE ON STATIC IMAGE DATASETS

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S1 [13], 2025 (Tipare et al.)	Static RGB hand gesture images → speech	2D-CNN gesture classifier + TTS synthesis	Custom static hand gesture image dataset under varied lighting	Degrades in poor illumination; cannot model dynamic multi-frame signs
S2 [17], 2022 (Sosa-Jiménez et al.)	Static and isolated Mexican SL images (healthcare)	Deep CNN classifier + speech synthesis	Custom Mexican SL primary-care gesture dataset	Restricted clinical vocabulary; validated only in controlled indoor settings
S3 [14], 2024 (Mahendran & Mahendran)	Static RGB images, ISL alphanumerics	Deep 2D-CNN and transfer-learning classifiers	Custom ISL alphanumeric static image set (A–Z, 0–9)	Limited to static fingerspelling; needs uniform background segmentation
S4 [21], 2025 (Potey et al.)	Static RGB gesture images (REST microservice)	2D-CNN wrapped as cloud REST microservice + audio output	Custom Divyang static sign image dataset	Network round-trip latency; needs stable internet connectivity
S8 [22], 2025 (Bharathi et al.)	Static RGB sign images + speech (bidirectional)	2D-CNN + NLP parser + ASR + TTS pipeline	Custom static sign image and speech corpus	Frame-by-frame static classification; accuracy drops on multi-word sentences
S9 [15], 2022 (Patil et al.)	Static RGB images, ISL alphabets and words	Deep neural network (DNN / 2D-CNN) + TTS	Custom ISL static gesture image dataset	Cannot capture temporal motion trajectories or co-articulation
S10 [23], 2025 (Bidirectional ML)	Static RGB hand images, two-way translation	2D-CNN feature extractor + classical ML classifiers	Custom bidirectional static sign image dataset	Small vocabulary; no temporal boundary detection
S11 [24], 2023 (Portable SL System)	Static RGB images on embedded portable hardware	Lightweight 2D-CNN on embedded processor	Custom portable-camera static gesture image set	Limited on-board compute restricts throughput and class scalability
S13 [16], 2025 (Dutta et al.)	Static RGB images, ISL alphabets	Convolutional Vision Transformer (CvT)	Benchmark and custom Indian SL static image dataset	High computational cost and memory footprint; needs large training sets
Deshmukh et al. [46], 2024	Static RGB images, Marathi Sign Language (MSL)	VGGNet transfer-learning CNN	Self-collected static Marathi alphabet image dataset	High parameter count (VGG backbone); restricted to isolated static MSL characters
Wadhawan & Kumar [47], 2020	Static RGB images, ISL alphabets, digits and signs	Deep 2D-CNN with multi-configuration filter banks	35,000 static RGB images across 100 ISL static sign classes	Appearance-based; sensitive to signer skin tone and background shifts
Katoch et al. [48], 2022	Static RGB images, ISL alphabets and digits	SURF descriptors + Bag-of-Visual-Words + SVM and 2D-CNN	Custom static ISL dataset (36 alphanumeric classes)	Relies on skin-colour segmentation; fails on cluttered backgrounds
Sharma et al. [49], 2021	Static RGB images, ISL alphabets and numbers	Pre-trained VGG16, ResNet50 and MobileNetV2 benchmark	Public static ISL alphanumeric image datasets	Evaluates only isolated single-frame postures; no temporal modelling
Giri et al. [50], 2024	Static hand landmarks from RGB images, MSL	MediaPipe Hands keypoint extraction + dense neural classifier	Custom static Marathi Sign Language character dataset	Classifies single-frame postures only; no temporal buffer or Idle state

Table II synthesises studies that operate on dynamic image sequences and multi-frame skeletal landmark trajectories, capturing temporal hand movement across isolated dynamic signs and sequence translations.

TABLE II. SUMMARY OF LITERATURE ON DYNAMIC IMAGE AND GESTURE SEQUENCE DATASETS

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S5 [26], 2025 (Comparative AI/ML)	Static and dynamic gesture sequences (review)	Comparative evaluation of CNN, RNN, LSTM, SVM, Random Forest	Multiple public static and dynamic SLR benchmark datasets	High annotation cost; no unified continuous evaluation
S12 [27], 2023 (Multimodal NMT)	Dynamic multimodal sign sequence representations	Transformer-based Neural Machine Translation (NMT)	Multimodal dynamic sign-to-text translation corpora	Quadratic self-attention complexity; unsuitable for low-power CPU edge devices
S14 [25], 2024 (Bagane et al.)	Dynamic speech-to-sign sequence mapping	ASR + NLP syntactic parsing + dynamic sign sequence lookup	Parallel speech-text and dynamic sign sequence corpus	Unidirectional (speech-to-sign); vulnerable to upstream ASR word errors
Lugaresi et al., Zhang et al., Bazarevsky et al. [6]–[8], 2019–2020	Dynamic 3D hand and body skeletal landmark sequences	MediaPipe Holistic, MediaPipe Hands (21 keypoints), BlazePose	Real-time multi-person hand and body pose tracking benchmarks	Raw image-space coordinates vary with camera distance; self-occlusion dropouts
Yan et al. [31], 2018	Dynamic 2D/3D skeletal keypoint graph sequences	Spatial-Temporal Graph Convolutional Network (ST-GCN)	Kinetics-Skeleton and NTU-RGB+D dynamic action datasets	Designed for gross whole-body actions, not fine-grained manual articulation
Mittal et al. [33], 2019	Dynamic 3D skeletal hand trajectory sequences	Modified 2-layer LSTM with reset gates for sequence segmentation	942 dynamic ISL sentences and 35 isolated dynamic words (Leap Motion)	Depends on dedicated external Leap Motion infrared sensor hardware
Kothadiya et al. [51], 2022	Dynamic frame sequences, isolated ISL words	DeepSign hybrid stacked LSTM and GRU sequence architecture	IISL2020 dataset (11 dynamic ISL word classes, video frame sequences)	Small vocabulary (11 dynamic signs); no idle rejection for continuous streams
Shenoy et al. [52], 2018	Dynamic hand gesture image sequences, ISL	Object stabilisation + face-hand centroid tracking + HMM and LSTM	Custom dataset of 33 static poses and 12 dynamic ISL hand gestures	Needs skin segmentation and face detection; sensitive to rapid illumination shifts
Nurtanio & Syafaruddin [53], 2023	Dynamic MediaPipe hand keypoint sequences	MediaPipe Hands (21 3D landmarks) + modified multi-layer LSTM	Custom dynamic sign language sequence dataset	Unidirectional LSTM lacks future context in the sliding window; no wrist normalisation
Gonçalves et al. [54], 2025	Dynamic MediaPipe holistic landmark sequences	Spatial-temporal Transformer encoder over skeletal keypoints	Dynamic sign language keypoint sequence corpus	Transformer attention overhead without local Conv1D inductive bias; no debounce gate
Joshi et al. [55], 2024	Dynamic skeletal landmark sequences, ISL	Enhanced LSTM with custom temporal attention mechanism	Custom dynamic Indian Sign Language gesture dataset	Evaluated on pre-segmented clips; forced softmax output on idle transitions

TABLE III. SUMMARY OF LITERATURE ON VIDEO-BASED CONTINUOUS SIGN LANGUAGE RECOGNITION

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S6 [18], 2025 (Abeje et al.)	Raw RGB video streams, Ethiopian SL words	Hybrid 2D-CNN spatial extractor + LSTM temporal classifier	Self-collected Ethiopian SL word-level video dataset	High per-frame CNN compute; needs pre-segmented word video clips
S7 [19], 2024 (Video SL Framework)	Continuous RGB video streams	Statistical spatio-temporal features + supervised ML classifiers	Custom continuous sign language video dataset	Highly sensitive to background motion clutter and ambient lighting changes
S15 [20], 2025 (Mosleh et al.)	RGB video streams, bidirectional Arabic SL	Hybrid deep CNN-RNN + fuzzy string matching	Arabic Sign Language continuous and word-level video dataset	Heavy GPU compute and memory footprint; needs large annotated video corpora
Koller et al. [3], 2015	Continuous RGB sign language video	Hybrid CNN appearance modelling embedded in statistical HMMs	RWTH-PHOENIX-Weather multi-signer continuous German SL video	Restricted to weather broadcast domain; complex multi-stage offline expectation-maximisation

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
Camgoz et al. [4], 2018	Continuous RGB video → spoken-language text	2D-CNN + attentional encoder–decoder Neural Machine Translation	RWTH-PHOENIX-Weather 2014T (8,257 continuous German SL videos)	Offline full-sentence processing; high word error rate out of domain
Cui et al. [28], 2019	Continuous RGB video sequences	2D-CNN + stacked BiLSTM with iterative CTC alignment refinement	RWTH-PHOENIX-Weather 2014 and Chinese Continuous SL video corpora	Needs multi-epoch iterative full-video decoding; not suited to live streaming
Camgoz et al. [29], 2020	Continuous RGB video (joint SLR + SLT)	Sign Language Transformer with joint CTC recognition and translation loss	RWTH-PHOENIX-Weather 2014T continuous video benchmark	Processes entire unsegmented sequences offline; heavy for CPU edge deployment
Zhou et al. [30], 2020	Continuous RGB video + multi-cue patches	Spatial-Temporal Multi-Cue (STMC) network + BiLSTM + CTC	RWTH-PHOENIX-Weather 2014 / 2014T and CSL continuous video	Multi-branch crop extraction (hand, face, full frame) imposes high GPU latency
Li et al. [10], 2020	Word-level RGB video clips, ASL	2D-CNN+GRU, 3D-CNN (I3D) [32], and pose-based TGCN baselines	WLASL benchmark (2,000 ASL words, 21,083 videos, 119 signers)	Isolated word-level clips only; no continuous sentence boundary handling
Sridhar et al. [9], 2020	Word-level RGB video clips, ISL	CNN-BiLSTM and OpenPose [34] keypoint sequence baselines	INCLUDE benchmark (263 ISL word classes, 4,287 HD videos)	Isolated ISL lexical items recorded in controlled studio settings; not MSL
Zuo & Mak [56], 2022	Continuous RGB video streams	C2SLR: 2D-CNN + BiLSTM + CTC with visual-gloss consistency constraints	PHOENIX-2014, PHOENIX-2014T and CSL continuous video datasets	Full-frame ResNet backbone needs GPU acceleration; offline sentence decoding
Hu et al. [57], 2023	Continuous RGB video streams	Self-Emphasizing Network (SEN) with spatial-temporal attention + BiLSTM-CTC	PHOENIX-2014, PHOENIX-2014T, CSL and CSL-Daily video corpora	Processes dense RGB video tensors; no explicit mechanism for live idle-state rejection
Hu et al. [58], 2023	Continuous RGB video streams	CorrNet: frame-wise trajectory correlation module + 2D-CNN + BiLSTM-CTC	PHOENIX-2014, PHOENIX-2014T, CSL and CSL-Daily video benchmarks	Dense spatial-correlation tensor computation raises memory and latency on CPU
Zuo et al. [59], 2023	Word-level and continuous RGB video + keypoint heatmaps	NLA-SLR: two-stream 3D-CNN (S3D) on RGB video and human keypoint heatmaps	WLASL, MSASL, NMFs-CSL and PHOENIX-2014 video benchmarks	Dual-stream 3D-CNN backbone is computationally prohibitive for edge devices
Cui & Zhang [60], 2023	Continuous RGB video streams	SLRFormer: Vision Transformer spatial-temporal encoder + CTC decoding	Continuous sign language video benchmark corpora	High self-attention memory footprint; no real-time sliding-window idle filtering

5.3 Modality Trends: RGB versus Skeleton

As Tables I–III show, most of the 15 primary studies use dense RGB image or video pipelines, with CNN benchmarks for static ISL/MSL images appearing in the domain literature [46]–[49] and multi-stream continuous video pipelines [56]–[60]. Landmark-based MediaPipe pipelines have been applied to static MSL postures [50] and to pre-segmented dynamic signs [53]–[55]. However, no study in the corpus combines zero-pixel, wrist-normalised landmark extraction with continuous idle-state rejection.

RGB (appearance-based) input. Raw frames contain handshape detail, facial expression and mouth patterns. These non-manual markers carry grammatical information in many sign languages [2]. A 3D convolutional network such as I3D [32] can learn motion and appearance together. The price is sensitivity to nuisance factors. The reviewed studies report degradation under poor lighting [13], sensitivity to background noise [19] and to lighting change [14], and a need for controlled backgrounds [14]. These models also require significant computing power and, because they process faces and surroundings, raise privacy concerns when used in hospitals or offices.

Sensor-based input. Glove- and depth-sensor-based systems, such as the modified LSTM on Leap Motion data [33], can be accurate but need specific hardware. They were excluded from the main corpus (EC3) because they cannot scale to a general audience.

Skeletal (landmark-based) input. Pose estimators such as OpenPose [34] and on-device estimators such as MediaPipe [6]–[8] convert each frame into a small vector of body and hand keypoints. Appearance is largely ignored, so the representation is low-dimensional, independent of clothing, skin colour, background and lighting, and much cheaper to store and process. Reference [31] shows that landmark sequences contain enough information to learn action recognition, and [30] shows that multi-cue sign recognition networks use RGB and pose cues together. The restrictions are also clear. A hands-only landmark stream carries no facial information. Trackers can be lost during hand-to-hand overlap or fast motion. Raw landmark coordinates are in image space and therefore vary with the signer's position and distance from the camera unless they are normalised. Table IV compares the two families.

TABLE IV. COMPARISON OF RGB AND SKELETAL INPUT FOR SIGN RECOGNITION

Criterion	RGB (appearance)	Skeletal landmarks	Evidence
Input size per frame	Full image (high-dimensional)	Tens to hundreds of coordinates	[6]–[8]
Sensitivity to lighting and background	High	Low (appearance discarded)	[13], [14], [19]
Inference cost	High, especially for 3D CNNs and Transformers	Low; suited to CPU and edge devices	[16], [18], [32]
Privacy	Faces and surroundings processed	No pixels need to be stored	Design property
Non-manual markers (face, mouth)	Available	Absent unless face landmarks are added	[2]
Main failure modes	Illumination, clutter, occlusion	Tracking dropout, camera-framing bias	[6]–[8]

5.4 Architectural Evolution

The corpus shows a progression from frame classifiers to sequence models (Fig. 2). Static systems map one image to one label using CNNs or DNNs [13]–[15]. The Convolutional Vision Transformer [16] is more accurate than traditional CNNs on Indian Sign Language images but needs high-end hardware and large training sets. Dynamic systems model time. Hybrid CNN–LSTM models are used for Ethiopian sign video [18] and, together with fuzzy matching, for Arabic Sign Language [20]. The CNN-plus-recurrent pattern used in earlier continuous recognition work [28] was adapted to CTC training. Later, Transformer sequences were used to recognise and translate jointly [29], but this requires working on the entire sequence, which is undesirable for real-time use.

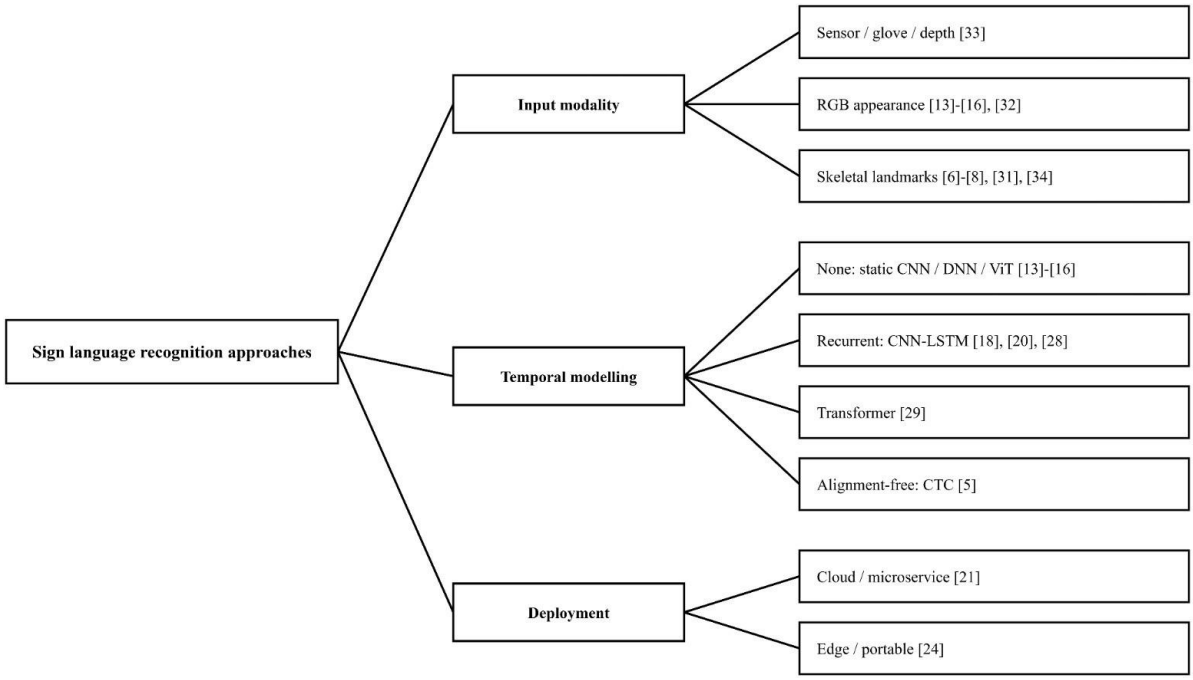


Fig. 2. Taxonomy of sign language recognition approaches by input modality, temporal modelling and deployment.

Three building blocks recur in the sequence-modelling literature and can be applied to landmark inputs. First, Long Short-Term Memory (LSTM) networks [35] model long temporal dependencies, and bidirectional recurrent networks [36] read both past and future context in a window. Second, attention mechanisms [37] let the model weight the frames that matter most for the discriminative movement, which is useful when only some frames in a window are relevant. Third, one-dimensional temporal convolutions [38] extract local motion patterns efficiently and in parallel. Isolated and continuous benchmarks and comparative evaluations of deep-learning methods [39] exist for both settings and provide a stable baseline for evaluating such combinations. We did not find any study in the primary corpus that integrates temporal convolution, bidirectional recurrence and attention over landmark sequences.

5.5 Continuous Recognition, Segmentation and Temporal Hallucination

In continuous recognition the start and end of each sign must be decided. There are two families of solutions. Alignment-based methods, such as CTC-trained models [5], [28], [29], do not need frame labels to learn the sentence-level mapping, and recent work adds visual alignment constraints to strengthen the feature extractor [40]. These techniques need sentence corpora with glosses and typically operate on an entire video. Window-based methods are the natural choice in practice because they classify short sliding windows, but they require an explicit representation of “no sign”.

Three studies in the primary corpus address dynamic signing [18]–[20]. Their results are good but imperfect: they recognise continuous word gestures [18], they are sensitive to background noise and lighting [19], and a related bidirectional system captures simpler sentences with higher accuracy [22]. None of the extraction records contains a mechanism or metric for rejecting resting or ambient movement. We use the term **temporal hallucination** for a confident prediction of a sign from input that contains no sign, such as resting hands, posture adjustment or transitions between signs. A closed-set softmax classifier must assign probability to its classes, so without a background class it will always output some sign. Softmax confidence in modern neural networks is also known to be miscalibrated [41] and should therefore be tested empirically rather than assumed.

5.6 Dataset Landscape for Regional Sign Languages

Fourteen of the 15 primary studies use a self-collected custom dataset; the exception is the comparative review [26], which draws on several benchmark datasets. This limits reproducibility and makes it hard to compare results across studies. Large benchmarks do exist for other languages: WLASL for ASL, with about 2,000 glosses [10]; INCLUDE for ISL word-level signs [9]; and RWTH-PHOENIX-Weather 2014T for continuous German Sign Language with spoken-language translations [4].

We identified three publicly available resources for MSL. None is a benchmark; all are repository entries. The first is a multimodal ISL dataset of English and Marathi alphabets and digits, recorded by 4 signers against a uniform background, with pose and skin-segmented representations, published in August 2026 [42] and containing 79 classes. The second is a Hugging Face dataset of static images for 43 classes of the Marathi alphabet [43]. The third is a Mendeley dataset of Indian Sign Language sentences with video and gloss pairs, where the gloss is in Marathi and follows subject–object–verb order [44]. Signer diversity, recording conditions, splits and annotation procedures were taken from the repository descriptions and were not independently verified. Together they give adequate coverage of static fingerspelling but do not form a benchmarked, multi-signer, continuous MSL corpus. Data scarcity is likewise identified as a major challenge in a survey of sign language machine translation [45].

5.7 Quality Assessment Results

Study-wise scores are given in Table V. No study was excluded: all 15 scored 8 or above (mean 9.0/10). Four studies scored 10 [16], [20], [22], [26], seven scored 9 and four scored 8. Objective clarity (QA1), method description (QA2) and discussion of limitations (QA5) scored 2 in every study. The mean scores for QA3 (data description) and QA4 (quantitative evaluation) were 1.53 and 1.47. Evaluation reporting is the weakest dimension: eight studies received a QA4 score of 1 because their results were presented qualitatively (for example “high accuracy” or “low latency”) rather than numerically.

TABLE V. STUDY-WISE QUALITY ASSESSMENT SCORES (MAXIMUM 10)

Study	QA1	QA2	QA3	QA4	QA5	Total
S1 [13]	2	2	2	1	2	9
S2 [17]	2	2	1	1	2	8
S3 [14]	2	2	2	1	2	9
S4 [21]	2	2	1	1	2	8
S5 [26]	2	2	2	2	2	10
S6 [18]	2	2	2	1	2	9
S7 [19]	2	2	1	1	2	8
S8 [22]	2	2	2	2	2	10
S9 [15]	2	2	1	2	2	9
S10 [23]	2	2	1	2	2	9
S11 [24]	2	2	1	2	2	9
S12 [27]	2	2	1	1	2	8
S13 [16]	2	2	2	2	2	10
S14 [25]	2	2	2	1	2	9
S15 [20]	2	2	2	2	2	10

6. Results and Findings

6.1 Architectures (RQ1)

The main corpus (S1–S15), complemented by the wider repository of static (Table I), dynamic-sequence (Table II) and continuous-video (Table III) literature, shows an architectural trajectory of three generations. Of the 15 primary studies, 5 focus on still-image gesture classification [13]–[17], 3 on dynamic or continuous video interpretation [18]–[20], 5 on bidirectional communication pipelines [21]–[25] and 2 on cross-cutting algorithmic or translation analysis [26], [27]. Across all 40 studies in the synthesis of Tables I–III, 14 use static single-frame images, 11 use dynamic gesture or landmark sequences and 15 use continuous or word-level RGB video.

The first generation (Table I) uses 2D convolutional networks, either custom shallow ConvNets [13]–[15], [21] or heavy transfer-learning backbones such as VGG16, ResNet50 and MobileNetV2 [46], [49], as spatial feature extractors over single RGB images. Dutta et al. [16] show that static ISL postures can be classified more accurately with CvT than with conventional 2D-CNNs, but CvT is essentially memoryless and cannot learn motion trajectories, velocity profiles or transitions between signs.

The second generation (Table II) introduces explicit temporal modelling over multi-frame trajectories or skeletal landmarks. Unidirectional LSTM and GRU stacks [33] classify isolated dynamic gestures by propagating hidden states forward in time, while Spatial-Temporal Graph Convolutional Networks (ST-GCN) [31] and landmark Transformers [54] model joint topology across frames. Even so, unidirectional recurrent units are prone to recency bias and have no look-ahead context in a sliding window, and pure Transformer encoders [27], [54] lack local inductive bias in short landmark windows and carry $O(T^2)$ self-attention cost. Adding custom temporal attention over recurrent hidden states [55] further improves frame weighting for dynamic ISL words, which supports the idea that frames in a gesture window do not all carry equal discriminative weight.

Continuous video-based models combine deep spatial backbones (2D-CNN, 3D-CNN/I3D [10], [32], S3D [59] or Vision Transformers [29], [60]) with sequence encoders (BiLSTM or Transformer) trained end to end with CTC loss [5], [28]–[30], [40], [56]–[58] (Table III). Recent approaches such as C2SLR [56], SEN [57] and CorrNet [58] use visual-gloss consistency and inter-frame correlation to reach low word error rates on RWTH-PHOENIX-Weather 2014T. Importantly, these video architectures are optimised for offline decoding of full sentences on high-end GPUs. For low-latency deployment on edge CPUs, a compact alternative to heavy 3D-CNN and full-sequence Transformer pipelines can be built from 1D temporal convolutions that extract local velocity and acceleration primitives in parallel, a bidirectional LSTM that captures onset and offset context within a fixed sliding window, and multi-head temporal attention [37].

6.2 Feature Extraction (RQ2)

The comparison of appearance-based RGB representations (Tables I and III) with skeletal landmark representations (Tables II and IV) shows a clear trade-off between the visual richness of appearance and robustness in deployment. All 15 primary studies, and the majority of continuous video benchmarks [3], [4], [9], [10], [28]–[30], [56]–[60], process dense pixel arrays. A standard 224 × 224 RGB frame contains 150,528 scalar values, or 4,515,840 values over a 30-frame window, so the early layers of 2D/3D CNNs and Vision Transformers spend many floating-point operations (FLOPs) filtering out lighting gradients, shadows, background noise, clothing and skin-colour variation. Accordingly, accuracy drops under non-uniform lighting [13], [19], complex backgrounds [14], [48] and cross-environment testing [17], [47] are reported repeatedly, for both static images and video.

By contrast, on-device skeletal estimators such as MediaPipe Hands and Holistic [6]–[8] reduce each frame to 21 3D hand points per hand. For two hands this is a vector of only 126 dimensions per frame,

or 3,780 values over a 30-frame window. That is a reduction of about 99.9% in input dimensionality per window compared with raw RGB frames (3,780 / 4,515,840 ≈ 0.08%). Dynamic SLR studies using MediaPipe keypoints [50], [53]–[55] indicate that skeletal streams are inherently robust to ambient light, background noise and signer appearance, that CPU inference on skeletal streams is feasible in real time, and that privacy is preserved because raw facial and environmental imagery is discarded.

However, we identify three issues in current skeleton pipelines [50], [53], [54] that must be overcome for reliable deployment. (i) Raw MediaPipe keypoint coordinates are normalised to the camera view, so they change whenever the signer moves horizontally or vertically or changes distance from the webcam. (ii) Single-hand dropout during fast movement, or a hand leaving the camera frame, changes the size of the input vector unless it is handled deterministically. (iii) A hands-only keypoint stream contains no non-manual facial markers [2]. Deterministic zero-fill padding together with anchoring each hand's coordinates to the wrist joint (landmark 0) and scaling by the palm length (wrist to middle-finger MCP, landmark 9) gives immunity to spatial framing and dropout without losing the zero-pixel computational benefit.

6.3 Datasets and Evaluation (RQ3)

RQ3 shows significant fragmentation in dataset construction, vocabulary scale and rigour of evaluation. Fourteen of the 15 primary studies (93.3%) evaluate on private, self-collected datasets that are not publicly available; only the comparative review [26] cross-references public benchmarks. The quality assessment also shows that two of the weakest aspects of the primary corpus are evaluation reporting, where 8 of 15 studies lack exact precision, recall or F1 figures, and dataset documentation, where 7 of 15 studies describe their data only partially (QA3 = 1).

There is a clear gap between high-resource international benchmarks and regional Indian languages. For continuous video research, the corpora used are RWTH-PHOENIX-Weather 2014T [4] (8,257 German SL sentences, 1,066 sign glosses, 9 signers) and the Chinese sign language corpora [28], [30], [56]–[58]. By contrast, small static alphanumeric datasets [14], [42]–[50] or small isolated dynamic word sets (such as the 11-word IISL2020 dataset [51] or a 12-gesture set [52]) are mainly used in studies of Indian and Marathi Sign Language.

For MSL, existing datasets have three major drawbacks. (i) They are restricted to static Devanagari alphabets and digits [42], [43], [46], [50], which reduces real communication to slow character-by-character fingerspelling instead of natural lexical signing. (ii) The 79-class Mendeley static dataset has little signer and environment variety, with only 4 signers recorded against a uniform background [42]. (iii) The repository uploads [43], [44] have no verified signer-independent train/validation/test splits and no continuous boundary annotations. Evaluating on random frames or clips from the same signers as the training data lets subject-specific hand geometry leak into the test set and inflates accuracy compared with a true signer-independent evaluation.

6.4 Real-Time Deployment and Hallucination (RQ4)

RQ4 reveals three operational barriers between laboratory SLR models and real-world use in clinics, classrooms and public offices: hardware compute bottlenecks, network dependency and continuous-stream hallucination. First, state-of-the-art video and Transformer architectures [16], [18], [20], [27], [29], [30], [56]–[60] need discrete GPUs to sustain real-time frame rates (25–30 FPS or more). On embedded or portable CPU hardware without skeletal dimensionality reduction, models are very sensitive to frame drops and latency increases [24]. Second, delegating inference to cloud microservices [21] adds network round-trip latency, creates a bandwidth problem for continuous video streaming, and fails in low-connectivity environments such as rural or public-sector sites.

Third, and most importantly, none of the 15 primary studies or the accompanying dynamic landmark studies [50]–[55] reports an engineering approach or quantitative metric for rejecting non-signing intervals during live continuous camera streaming. In real conversation a signer's hands may rest for long periods, for example while repositioning, adjusting posture, listening to the other person, or making transitional co-articulations between glosses. A sliding-window classifier trained only on K sign classes and fed a continuous 30 FPS stream must, because of its softmax output layer, distribute all of its probability mass over those K classes. The system therefore keeps emitting spurious sign tokens (temporal hallucination), even for out-of-distribution input [41].

We propose that this can be suppressed without sacrificing responsiveness through a multi-layered defence: (i) training an explicit (N + 1)-th Idle sink class covering resting postures, empty-frame zero tensors, self-adaptors and inter-sign transitions; (ii) enforcing a calibrated posterior probability gate ($\tau = 0.85$) validated by Expected Calibration Error (ECE) [41]; (iii) requiring temporal persistence across $k = 3$ consecutive sliding windows to filter transient transition spikes; and (iv) applying a post-emission cooldown lock ($T_{cool} = 15$ frames) to prevent duplicate token emission while a sign is held. These are design proposals; they are not validated in this review.

6.5 Threats to Validity

Several threats to validity were considered and reduced. For construct validity, searches used broad terms for assistive translation, specific architectural terms and specific landmark terms across eight databases (Appendix A). Replication strings were added for Scopus and the ACM Digital Library (Section 3.3), and forward and backward snowballing [3]–[10], [28]–[34] captured the foundational skeleton-based and continuous SLR literature, while the primary set covers 2022–2026. For internal validity, study selection and the five-point quality scoring (QA1–QA5) were carried out by the project team using pre-developed rubrics (Section 3.6). The criteria were explicit, which limited subjective bias, although external double-blinding was not possible. For conclusion and external validity, raw accuracy percentages were not pooled across the 14 of 15 primary studies that use private data with heterogeneous metrics. The synthesis is instead organised by input modality (Tables I–III) and by architectural complexity and deployment failure (Tables IV and V). Finally, the metadata of the emerging MSL datasets [42]–[44] is marked as unverified until validated under the proposed validation agenda.

7. Research Gaps

The cross-modality synthesis in Sections 5 and 6 reveals four basic research gaps (G1–G4) that existing sign language recognition models cannot overcome if they are to become a workable, real-time assistive tool for Marathi-speaking deaf people. Evidence, practical implications and research opportunities for each gap are summarised in Table VI.

7.1 Gap 1 (G1): Computational overhead and edge-hardware incompatibility. The main obstacle to real-world adoption is the high computational cost of pixel-based deep learning. Prevalent SLR models include deep 2D CNNs (VGGNet [46], ResNet50 [49], DenseNet), 3D spatio-temporal convolutions (I3D [10], [32], S3D [59]), hybrid CNN-LSTM models ([18], [20], [28]) and Vision Transformers (CvT [16], Sign Language Transformers [29] and SLRFormer [60]), as detailed in Tables I and III. A 30-frame window of 224 × 224 RGB images contains more than 4.5 million input values, most of which describe background texture, clothing folds and lighting gradients rather than hand kinematics. These models perform well on benchmarks run on workstation GPUs, but they are severely limited by latency or thermal throttling on commodity CPU-only laptops, budget smartphones and portable embedded systems [24]. Delegating heavy pixel models to cloud microservices [21] adds network latency and does not work in low-connectivity hospitals, rural schools or public offices. A

practical assistive engine must drastically reduce input dimensions and parameters so that it can run locally on standard CPU hardware at a continuous 30 FPS.

7.2 Gap 2 (G2): Spatial bias, camera-framing sensitivity and domain shift. Models trained in controlled laboratory conditions often fail when used with new rooms, webcams and users, and this happens in two ways. First, appearance-based RGB classifiers [13]–[15], [17], [19], [46]–[48] learn the illumination profile, background colour and signer skin tone of the dataset and misclassify data with slight changes in light or background. Second, existing skeletal pipelines that use MediaPipe or OpenPose [6]–[8], [34], [50], [53], [54] generally feed raw landmark coordinates in image space, $(x, y, z) \in [0, 1]^3$, to the classifier, which causes severe camera-framing bias. Raw viewport coordinates encode where the hand is in the frame and how far the signer is from the lens, so the numeric distribution changes sharply even when the physical sign is unchanged, for example if the signer steps half a metre back, leans sideways or uses a webcam with a different field of view. Skeletal models are therefore fragile in real camera setups because they lack frame-by-frame geometric normalisation with explicit translation and scale invariance.

7.3 Gap 3 (G3): Temporal hallucinations and absence of idle-state rejection. A major limitation of the reviewed literature is the disconnect between isolated laboratory testing and live inference. Most dynamic and video-based studies in Tables II and III ([18]–[20], [51]–[55]) train and evaluate on pre-segmented clips that are assumed to contain exactly one valid sign. In spontaneous interaction, signing is frequently interrupted by idle hands in the lap or at the side, by self-adaptors such as scratching or adjusting glasses, by hands leaving the camera view, and by fast co-articulation between signs. A sliding-window classifier with a closed-set softmax output over N classes, fed a live webcam stream, is mathematically forced to place 100% of its probability mass on those N classes at every window. It therefore keeps predicting signs confidently during silence and ambient motion, a problem made worse by the over-confidence that modern deep networks show on out-of-distribution inputs [41]. None of the 15 primary studies explicitly uses an Idle sink class, a confidence-calibration analysis or multi-window debounce filtering, and none reports false activations per minute during non-signing periods.

7.4 Gap 4 (G4): Regional data scarcity for continuous MSL. Unlike American Sign Language (WLASL [10]) and German Sign Language (RWTH-PHOENIX-Weather 2014T [3], [4]), which have large multi-signer video benchmarks, MSL has severe resource scarcity. The publicly available MSL datasets [42], [43], [46], [50] are almost entirely limited to static single-frame images of the Devanagari manual alphabet and digits, recorded by few participants (for example just four signers against a plain background in [42]). The only repository entry with sentence-level Marathi glosses [44] is not peer-reviewed, has no signer-independent partitions or idle/transition class, and is unverified. Because of regional lexical markers and subject–object–verb (SOV) syntax, models trained on ASL or on static fingerspelling data cannot be expected to generalise to natural continuous Marathi signing. A documented, multi-signer continuous MSL dataset with strict signer-independent splits and an explicit Idle class is therefore essential for reproducible regional assistive technology.

TABLE VI. RESEARCH GAPS, EVIDENCE AND OPPORTUNITIES

Gap	Evidence from literature	Why it matters	Research opportunity
G1. Computational overhead. Pixel-based models are too heavy for edge devices.	Hybrid CNN-LSTM, Vision Transformer and NMT approaches report heavy compute or data demands [16], [18], [20], [27]; portable systems report performance limits from scarce resources [24]; 3D CNNs process full frames [32].	Assistive technology must run on ordinary laptops and phones to reach the community it is meant for.	Replace pixel input with landmark vectors and compare model size, latency and frame rate on CPU-only hardware.

Gap	Evidence from literature	Why it matters	Research opportunity
G2. Spatial bias and domain shift. Models fail on real-world hardware because of camera-framing bias.	Limitations cite lighting, background clutter, occlusion and camera-angle sensitivity [13], [14], [17], [18], [19]. Public MSL data were recorded by four participants against a uniform background [42]. Raw landmark coordinates depend on position and distance from the camera.	A model that works only on the recording setup of its training data fails for a different user, camera or room.	Normalise landmarks relative to a wrist anchor, and evaluate across signers, devices and camera positions.
G3. Temporal hallucinations. Systems cannot filter ambient noise and resting states.	No primary study records idle-state rejection or a false-activation metric (Section 6.4). A closed-set softmax must always pick a class [5], and its confidence is poorly calibrated [41].	A live system that outputs signs while the signer rests or moves naturally is unusable and erodes trust.	Add an explicit Idle class, a confidence threshold and temporal debouncing, and measure false activations per minute.
G4. Regional scarcity. There is no robust MSL dataset.	Public MSL resources are small and mostly static [42], [43]; a sentence-level video and gloss resource is unverified [44]. Benchmarks exist for ASL [10], ISL words [9] and German SL [4].	Without a shared multi-signer, continuous benchmark, MSL results cannot be compared or reproduced.	Collect and release a multi-signer MSL corpus with documented protocols, signer-independent splits and an Idle class.

8. Future Research Directions (Proposed Architecture)

8.1 Positioning

The four gaps point to a single design target: a camera-only engine that classifies signs from landmark sequences, runs on a CPU, rejects non-signing input and is trained on MSL data. This section describes such an engine as a design hypothesis grounded in the reviewed evidence. It has not been validated in this review, and the numerical settings below are initial values to be tuned.

8.2 Zero-Pixel Extraction with Root-Anchor Normalisation

Each webcam frame is passed to the MediaPipe Holistic [6] tracker, and only the two hand streams (21 3D landmarks each) are kept (Stage A of Fig. 3). No pixels are stored or transmitted; only landmark coordinates are retained and the frame is discarded. This is the zero-pixel property: it targets G1 and gives a privacy benefit. Each frame yields a 126-dimensional vector (2 hands × 21 landmarks × 3 coordinates). Additional pose landmarks [8] can be added as an extension to capture upper-body position.

To address G2, all landmarks are referenced to the wrist of their hand (the root anchor, landmark 0) and scaled by the distance from the wrist to the base of the middle finger (landmark 9), as in Eq. (1):

$$\tilde{p}_{t,h,j} = \frac{p_{t,h,j} - p_{t,h,0}}{\left\| p_{t,h,9} - p_{t,h,0} \right\|_2 + \varepsilon} \qquad (1)$$

Here $p(t,h,j) \in \mathbb{R}^3$ is the raw three-dimensional coordinate at frame $t \in \{1, \ldots, T\}$ for hand $h \in \{L, R\}$ and landmark index $j \in \{0, \ldots, 20\}$; $p(t,h,0)$ is the wrist root anchor; $p(t,h,9)$ is the middle-finger metacarpophalangeal (MCP) knuckle; $\|\cdot\|_2$ is the Euclidean norm; and $\varepsilon = 10^{-6}$ prevents division by zero. Under Eq. (1) the normalised coordinate is invariant to in-plane translation and to camera-to-signer distance scaling. When a hand is not detected in frame t, its 63 values are set to zero (zero-fill) before normalisation, so the 126-dimensional frame vector $\tilde{x}(t) \in \mathbb{R}^{126}$ keeps a fixed shape (a zero vector stays zero under Eq. (1)). One trade-off should be evaluated empirically: anchoring each hand independently discards inter-hand distance, which can be restored by appending a 3-dimensional inter-wrist displacement vector $\Delta p(t) = p(t,R,0) - p(t,L,0)$, or upper-body BlazePose landmarks [8].

8.3 Hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention Model

The classifier operates on a sliding window of T = 30 normalised frames, so its input is $X \in \mathbb{R}(30\times126)$ (Stage B of Fig. 3). Three stages are chosen for their specific functions. First, local short-range velocity and acceleration features $C \in \mathbb{R}(30\times128)$ are extracted in parallel along the time axis by two stacked one-dimensional temporal convolution (Conv1D) blocks [38] with kernel size 3, batch normalisation and ReLU activation. Second, C is processed in both forward and backward directions by a bidirectional LSTM [35], [36], and the two hidden states are concatenated to give context-enriched representations $H = [h(1), \ldots, h(30)] \in \mathbb{R}(30\times256)$, as in Eq. (2):

$$c_t = \mathrm{ReLU}(\mathrm{BN}(W_c * \tilde{x}_t + b_c))$$

$$h_t = \left[\overrightarrow{\mathrm{LSTM}}(c_t)\,;\, \overleftarrow{\mathrm{LSTM}}(c_t)\right] \quad (2)$$

Third, a multi-head temporal self-attention (MSA) module [37] with M = 4 heads (query/key dimension dk = 64 per head) computes scaled dot-product attention weights over the BiLSTM sequence H, so that frames containing the core manual stroke receive high weight and boundary-transition and rest frames receive low weight. This is followed by a residual connection, Layer Normalisation (LN), global temporal average pooling and an (N + 1)-way softmax classifier (Eq. (3)):

$$A_m = \mathrm{softmax}\left(\frac{Q_m K_m^{\top}}{\sqrt{d_k}}\right) V_m$$

$$\hat{y} = \mathrm{softmax}(W_o \cdot \mathrm{Pool}(\mathrm{LN}(H + \mathrm{Concat}(A_1, \ldots, A_M) W_p)) + b_o) \quad (3)$$

Here Qm, Km, Vm are the linear query, key and value projections of H for attention head m, and ŷ (with N + 1 entries, each in [0, 1]) is the posterior probability distribution over the N Marathi sign classes and the explicit Idle class (class index 0). The initial reference configuration is given in Table VII.

TABLE VII. INITIAL REFERENCE CONFIGURATION (ILLUSTRATIVE; TO BE TUNED)

Stage	Setting	Role
Input	30 × 126 tensor	Landmark window
1D-CNN	Conv1D, 64 and 128 filters, kernel 3, batch norm, dropout 0.3	Local temporal features
BiLSTM	128 units per direction, return sequences, dropout 0.3	Bidirectional context in the window
Temporal attention	4 heads, residual + layer norm, global average pooling	Frame weighting
Classifier	Dense layer 256 → N + 1, softmax	Sign or Idle

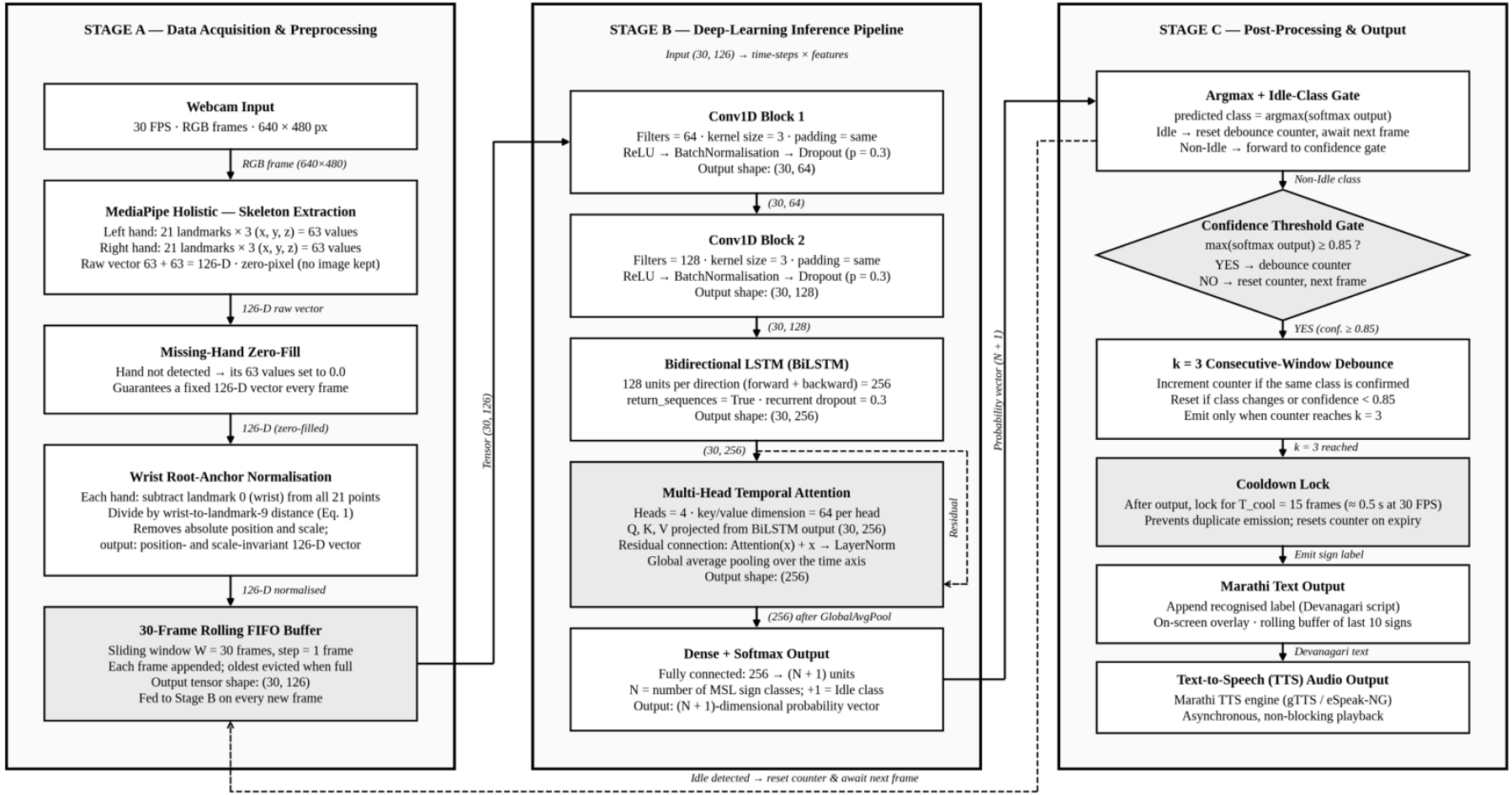


Fig. 3. Proposed camera-only system architecture: Stage A, acquisition and preprocessing; Stage B, hybrid 1D-CNN + BiLSTM + temporal-attention classifier; Stage C, post-processing and output. Dashed lines show the Idle feedback path and the attention residual connection.

8.4 Inference Control: Rolling Buffer, Idle Class and Confidence Debounce

G3 is handled at the level of the inference loop, not inside the network. Landmark vectors are held in a 30-frame (about 1 second) first-in-first-out (FIFO) buffer. An explicit Idle class is trained on resting hands, hands out of frame (all-zero windows), posture adjustment, transitions and other non-sign movement, so that "no sign" has somewhere to go instead of being forced onto a sign class. A prediction is accepted by the confidence-debounce filter only if the top class is not Idle, the top probability is at least 85%, and the same class is predicted for k consecutive windows (k = 3 as a starting value). Once a token is emitted, a cooldown lock (15 frames, or 0.5 s at 30 fps, as the initial value) prevents the same sign from being emitted again, and the debounce counter is reset when the lock ends. When Idle is detected, the lock is also released. The live workflow is in Fig. 4 and the sequence is summarised in Stage C of Fig. 3.

Let c(w) = argmax ŷ(w,i) and p(w) = max ŷ(w,i) be the predicted class and the maximum posterior probability at sliding window w. A sign c(w) is emitted at window w if and only if the debounce and confidence gate of Eq. (4) is true and w is outside the cooldown window Tcool:

$$E_w = \mathbb{1}(c_w \neq 0) \cdot \mathbb{1}(p_w \geq \tau) \cdot \prod_{i=0}^{k-1} \mathbb{1}(c_{w-i} = c_w) \cdot \mathbb{1}(w \notin T_{cool}) \tag{4}$$

Here □(·) is the indicator function, the confidence threshold is τ = 0.85, and the number of consecutive confirming windows is k = 3. Softmax confidence in deep networks can be miscalibrated [41], so τ should be calibrated on a held-out validation split [41] using a reliability diagram and Expected Calibration Error (ECE), and its sensitivity should be reported through a threshold ablation (Table VIII). The whole live inference control loop, ending in Marathi Devanagari text or text-to-speech (TTS) synthesis, is shown in Fig. 4.

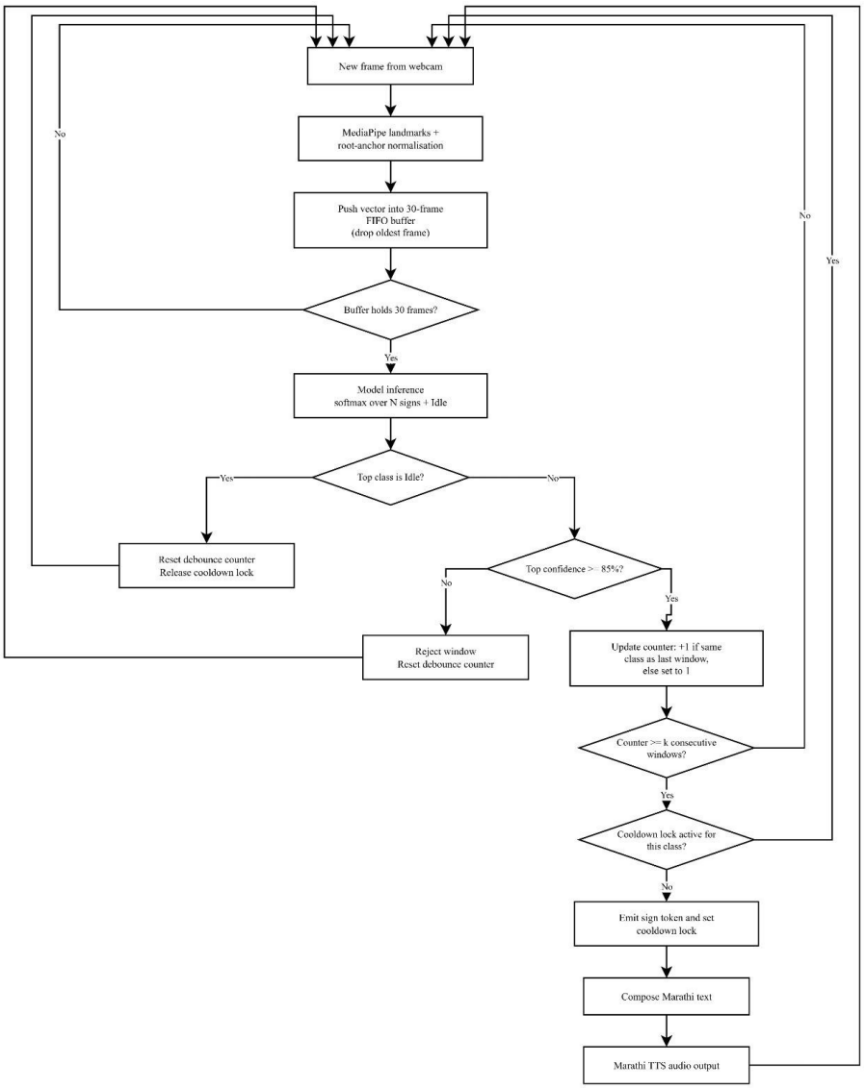


Fig. 4. Continuous live inference workflow.

8.5 Validation Agenda

The design should be evaluated against the gaps it addresses. The planned experiments and metrics are listed in Table VIII. Ablating each design element allows the contribution of the root anchor, the Idle class and the debounce filter to be measured rather than assumed.

TABLE VIII. VALIDATION PLAN LINKED TO RESEARCH GAPS

Gap	Design element	Planned validation and metrics
G1	Zero-pixel landmark input; compact hybrid model	Parameter count, model size, latency per window and sustained frames per second on CPU-only hardware, against an RGB baseline.
G2	Root-anchor normalisation	Signer-independent splits; tests across webcams, distances and framings; accuracy with and without normalisation.
G3	Idle class; 85% threshold; debounce	False activations per minute on long idle and ambient-motion recordings; expected calibration error; ablation of each component; threshold sweep.
G4	New multi-signer MSL corpus	Documented collection protocol, signer-independent train/test split, per-class accuracy and macro-F1, and an explicit Idle class.

Once this baseline is validated, further directions follow: adding face landmarks for non-manual markers, extending the system to produce sign-to-speech output, and testing it in realistic environments such as clinics and classrooms with deaf users.

9. Future Scope

The proposed zero-pixel hybrid 1D-CNN + BiLSTM + multi-head temporal attention architecture is a lightweight baseline for privacy-preserving continuous MSL recognition. Several high-impact extensions define the future direction of the BhashaBridge project.

9.1 Integration of Non-Manual Markers via Facial and Upper-Body Landmarks

Manual handshapes are accompanied by non-manual signals such as eyebrow raising or furrowing, mouthing, eye gaze and head tilt, which help distinguish declarative statements from questions, negations and conditional clauses in natural signing [2]. Future versions will append a small number of MediaPipe Face Mesh keypoints (16, covering eyebrows, lips and nose tip) and 6 BlazePose upper-body landmarks (shoulders and elbows) [6], [8] to the 42 hand landmarks (126 coordinates) saved at present, while keeping CPU throughput high. This fusion keeps the zero-pixel privacy guarantee and makes it possible to disambiguate identical manual trajectories that differ in facial grammar or body location.

9.2 Gloss-to-Marathi Grammatical Translation Using On-Device Language Models

In continuous sign language the output is a sequence of lexical glosses without inflection, whereas natural spoken and written Marathi is morphologically rich, with postpositions, verb agreement (gender, number, person) and tense markers. A key future goal of BhashaBridge is an offline, 4-bit quantised Small Language Model (SLM), or a fine-tuned IndicBART or mT5 sequence-to-sequence module, placed after the debounce filter, to reconstruct fluent, grammatically inflected Devanagari Marathi sentences before text-to-speech.

9.3 Bidirectional Communication via Speech-to-Sign 3D Avatar Synthesis

Two-way communication is essential for real accessibility in hospitals, banks and government offices. BhashaBridge will add a reverse pipeline that takes spoken Marathi from a doctor, teacher or clerk, converts it with offline Automatic Speech Recognition (ASR) into an SOV sign gloss, and uses that gloss to drive a lightweight parametric 3D skeletal avatar that shows continuous MSL animation with smooth co-articulation transitions.

9.4 Standardised Multi-Dialect Maharashtra MSL Benchmark Corpus

Sign vocabulary and lexical conventions differ between regions of Maharashtra, such as Pune/Western Maharashtra, Mumbai/Konkan, Vidarbha, Marathwada and North Maharashtra. Further data collection will involve deaf associations and special-education schools across Maharashtra to build an open-access MSL video benchmark of more than 500 lexical signs and 2,000 continuous SOV sentences from 50+ native signers, with standardised signer-independent partitions annotated with idle and transition boundaries.

9.5 Edge Quantisation, Mobile Packaging and Real-World Field Trials

To move from prototype to wide deployment, the trained hybrid model will be exported to ONNX Runtime and TensorFlow Lite with post-training INT8 weight quantisation, with the goal of sub-15 ms per-window inference on budget Android smartphones and standalone Raspberry Pi kiosk terminals. Word Error Rate (WER), False Activations Per Minute (FAPM) and System Usability Scale (SUS) scores will be measured in longitudinal field evaluations with deaf participants in primary-healthcare outpatient clinics (OPDs), municipal e-Seva kendras and inclusive classrooms at PCCoE.

10. Conclusion

To understand what is known, and what is missing, about real-time sign language recognition for Marathi Sign Language, a systematic search of eight databases, supplemented by skeleton-based and continuous-recognition literature, was conducted following the PRISMA guidelines. The main corpus consists of appearance-based CNN pipelines that are evaluated on private datasets in controlled settings. Hybrid CNN–LSTM and Transformer models improve sequence modelling but need significant compute, and the behaviour of a continuous system while the signer is not signing has not been reported.

Four gaps were identified: computational overhead, spatial bias and domain shift, temporal hallucination, and regional data scarcity. The proposed direction addresses each of them: zero-pixel landmark input, wrist-anchored normalisation, a hybrid 1D-CNN + BiLSTM + multi-head attention classifier, a 30-frame rolling buffer with an Idle class and a calibrated confidence debounce filter, and a purpose-built MSL corpus. The validation agenda in Section 8.5 (Table VIII) and the roadmap in Section 9 are intended to lead to an assistive tool that runs on an ordinary webcam and personal computer, needs no special equipment, and is robust to changes in lighting and camera position, so that Marathi-speaking deaf users have a practical communication tool in hospitals, classrooms and public offices. The performance of this design remains to be demonstrated experimentally.

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Appendix A – Search Log

The table records the per-database execution of the search strategy of Section 3.3 between 28 July and 5 August 2026. In total, 140 records were retrieved. After 18 duplicates were removed, 122 unique records were screened by title and abstract and 84 were excluded, mainly for publication before 2022, non-English language, absence of peer review, or dependence on non-camera hardware. Thirty-eight records were assessed in full text, and 23 were excluded for insufficient methodological detail, purely statistical approaches, or scope outside the four research questions, leaving the 15 studies (S1–S15) used in this review.

TABLE A1. DATABASE-WISE SEARCH LOG

Database	Date searched	Search string	Filters	Records
IEEE Xplore	28–31 Jul 2026	("sign language recognition" OR "gesture recognition") AND ("bidirectional translation" OR "sign-to-speech" OR "speech-to-sign") AND ("deep learning" OR "YOLO" OR "large language model")	2022–2026; English; Journal/Conf.	52
Google Scholar	28–31 Jul 2026	sign language recognition AND (bidirectional translation OR sign-to-speech OR speech-to-sign) AND (deep learning OR YOLO OR large language model)	2022–2026; English	30

Database	Date searched	Search string	Filters	Records
SpringerLink	1–3 Aug 2026	("sign language recognition") AND ("deep learning" OR "CNN-LSTM" OR "vision transformer") AND ("bidirectional" OR "assistive")	2022–2026; English; Article/Review	18
ScienceDirect	1–3 Aug 2026	("sign language") AND ("machine learning" OR "deep learning") AND ("translation" OR "communication system")	2022–2026; English	14
PubMed / PMC	3–4 Aug 2026	("sign language recognition") AND ("assistive technology" OR "hearing impaired") AND ("artificial intelligence")	2022–2026; English	10
Wiley Online Library	4 Aug 2026	("sign language recognition" OR "gesture recognition") AND ("deep learning" OR "neural network")	2022–2026; English	8
Nature (Scientific Reports)	4 Aug 2026	("sign language") AND ("recognition" OR "translation") AND ("AI" OR "deep learning")	2022–2026; English	5
Frontiers	5 Aug 2026	("sign language recognition") AND ("bidirectional" OR "assistive communication")	2022–2026; English	3
Total				140